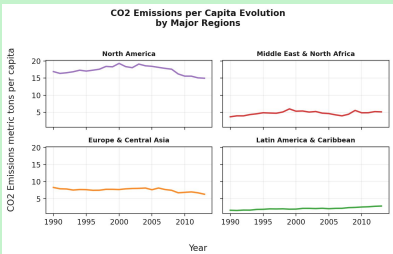
DV-Evol Bad Case vega-lite：Visual mapping failure

Question: Convert the 1990–2014 regional CO2 emissions data from mean to country-level median aggregation and replace the stacked area chart with a faceted line chart using specific HEX color mappings.

Gemini-3-pro-preview:

Ground Truth:



Plot code:

```
{
  "$schema": "https://vega.github.io/schema/vega-lite/v5.json",
  "data": {"url": "result.csv"},
  "facet": {"field": "Region", "columns": 2, "header": {"title":
null}},
  "spec": {
    "width": 300,
    "height": 200,
    "mark": {"type": "line", "point": true},
    "encoding": {
      "x": {"field": "Year", "type": "quantitative", "scale":
{"domain": [1990, 2014]}, "axis": {"format": "d"}},
      "y": {"field": "CO2", "type": "quantitative", "title": "Median
CO2"},
      "color": {
        "field": "Region",
        "scale": {
          "domain": ["North America", "Middle East & North
Africa", "Europe & Central Asia", "Latin America & Caribbean"],
          "range": ["#9467bd", "#d62728", "#ff7f0e", "#2ca02c"]
        },
        "legend": null
      }
    }
  },
  "resolve": {"scale": {"y": "independent"}}
}
```

Error Analysis: 1. Data Integrity

❌1.1 Trend/Pattern: 0/1 The distribution curves successfully mirror the multi-modal trends of the Ground Truth.

❌1.2 Completeness: 0/1 Critical data series or specific years are missing from the visualization.

❌1.3 Label Accuracy: 0/1 Axis tick values and labels do not precisely match the original data points.

❌1.4 Data Consistency: 0/1 Proportional relationships between income groups remain accurate across the X-axis.

2. Style Imitation

✅2.1 Color Palette: 1/1 The model correctly applied the designated HEX codes for all income categories.

❌2.2 Chart Elements: 0/1 Specific ridgeline outlines and gridline styles were not properly reproduced.

✅2.3 Background: 1/1 The background color and light/dark mode are consistent with the Ground Truth.

❌2.4 Font and Text Styles: 0/1 Font weight and sizing for titles and labels deviate from the original style.

❌2.5 Element Spacing: 0/1 Distances between labels and the plot area are uneven, creating a cluttered look.

❌2.6 Borders and Shadows: 0/1 Visual effects like drop shadows or specific border strokes were omitted.

3. Layout & Aesthetics

❌3.1 Labels: 0/1 Legend and title positioning are misaligned compared to the standard layout.

✅3.2 Readability: 1/1 Excessive ridge overlap and small font sizes compromise the overall legibility.

❌3.3 Axis Alignment and Spacing: 0/1 The axes are poorly calibrated, causing data to shift away from intended baselines.

❌3.4 Element Alignment: 0/1 Graphical components fail to align precisely with their respective axis ticks.

❌3.5 Proportionality and Aspect Ratio: 0/1 An incorrect aspect ratio has led to vertical distortion of the ridge shapes.

❌3.6 Overall Balance: 0/1 The composition lacks visual equilibrium, making the chart difficult to interpret quickly.

Score: 3/16